

Resource Bounds Without Substrate Identification: Exact Stochasticity and Substrate Skepticism

August 2026

Abstract

Substrate skepticism distinguishes the internal causal organization of a world from the physical system, if any, that realizes it. This note asks a narrower question: can internal stochastic structure constrain realizations even when it does not identify them? For finite controlled processes we separate two exact information costs. The *on-path cost* ρ_n is the worst, over intervention policies, of the Shannon entropy of the realized transcript. The *counterfactual seed cost* χ_n is the minimum entropy of a single policy-independent exogenous seed whose deterministic response tree reproduces the transcript law under every admissible policy. We prove $\rho_n \leq \chi_n$, exhibit strict inequality, characterize χ_n operationally, and derive a source-aggregation bound showing that deterministic layers cannot create the required common-seed entropy. We then relativize the cost to admissible realization classes and approximation tolerances and prove a quantitative total-variation lower bound showing that small statistical error cannot collapse the required information arbitrarily. We distinguish exact payment from computational concealment, isolate the additional consistency issue posed by infinite horizons, and derive a relative no-go theorem for closed deterministic hosts whose policy-independent initial source has bounded entropy. At infinite horizon a positive on-path entropy rate yields a sharp alternative: the required information must be stored in unbounded boundary data or supplied by continuing exogenous innovation. Only after adding terminality, resource-ancestry, finite-boundary, and ontic-bridge assumptions does this become a claim of fundamental stochasticity. The result is therefore deliberately limited: the costs can exclude resource-deficient realization classes and constrain the provenance of exact stochastic information, but they neither identify the actual substrate nor certify that the target world itself is fundamental. Substrate skepticism survives in a sharper, resource-theoretic form.

1 Introduction

The central claim of substrate skepticism is a separation rather than a simulation hypothesis:

$$\text{Obj}(W) \not\Rightarrow \text{Fund}(W). \quad (1)$$

The fact that an internally accessible world has persistent objects, stable causal regularities, public experimental structure, and reproducible responses to intervention does not by itself determine whether that world is fundamental or is realized by a distinct physical system [1]. In the formal version, two realizations are internally observationally equivalent when

$$R_1 \equiv_{\text{obs}} R_2 \iff \mu_{R_1}^{\pi,n} = \mu_{R_2}^{\pi,n} \text{ for every admissible policy } \pi \text{ and every finite } n, \quad (2)$$

where $\mu_R^{\pi,n}$ denotes the law of the internally accessible transcript. No test whose input is restricted to those transcripts can distinguish members of the same equivalence class.

There is, however, a different question. Even if internal evidence does not identify a realization, can it impose a lower bound on resources that any realization in a specified class must supply? Stochasticity makes this question natural: a deterministic mechanism can reproduce a stochastic interface only if some exogenous information is supplied somewhere, or if exact reproduction is weakened to an approximation relative to a restricted observer.

The relevant information-theoretic ingredients are not new in isolation. Minimum-entropy exogenous variables occur in entropic causal inference [2]; exact common randomness and exact channel synthesis quantify resources needed for exact generation [3, 5]; sequential channel synthesis studies temporally ordered synthesis tasks [6]; and abstract resource theories formalize conversion under a distinguished class of free operations [7]. The purpose of the present note is therefore not to claim priority for minimum exogenous entropy. It is to use these ideas to sharpen substrate skepticism, while making explicit several distinctions that are otherwise easy to blur:

- (i) randomness consumed on the realized path versus a common seed fixing all counterfactual branches;
- (ii) exact realization versus statistical or computational approximation;
- (iii) an abstract causal realization versus a physically admissible implementation;
- (iv) finite-horizon lower bounds versus a single consistent infinite-horizon realization;
- (v) a lower bound on realization resources versus a claim of fundamentality.

The main conclusion is correspondingly modest: *substrate non-identifiability is compatible with nontrivial substrate resource bounds.*

2 Controlled interfaces and two information costs

Fix finite action and observation alphabets $\mathcal{A}$ and $\mathcal{O}$. All Shannon entropies are discrete, measured in bits, and may take the value $+\infty$; no use is made of differential entropy. At time t , a finite internal history is

$$h_{t-1} = (a_1, o_1, \dots, a_{t-1}, o_{t-1}), \quad h_0 = \emptyset.$$

A deterministic intervention policy $\pi \in \Pi_n$ chooses $a_t = \pi_t(h_{t-1})$. A controlled world W is specified, through horizon n , by causal kernels

$$P_W(o_t \mid h_{t-1}, a_t), \quad 1 \leq t \leq n. \quad (3)$$

These kernels induce, for every deterministic policy π , a transcript

$$T_n^\pi = (A_1, O_1, \dots, A_n, O_n)$$

with law $\mu_W^{\pi, n}$.

Remark 1 (Why deterministic policies suffice). *A randomized adaptive policy can be represented by an auxiliary random seed U_π independent of the world. Conditional on $U_\pi = u$, all of its action choices are deterministic functions of the preceding history, so the resulting transcript law is a mixture of laws generated by deterministic policies. Consequently, equality of the policy-indexed interface for every deterministic policy implies equality for every externally randomized mixture as well. We therefore quantify over deterministic policies throughout. This also prevents randomness supplied by the experimenter's own randomizer from being charged to the stochastic resource of the world. If the randomizer is instead treated as an internal physical subsystem that may be correlated with the putative substrate, it belongs to an enlarged world model rather than to an independent randomized policy.*

2.1 On-path cost

The weakest exact information requirement concerns only the path that is actually realized.

Definition 1 (On-path exact cost). *Define*

$$\rho_n(W) := \sup_{\pi \in \Pi_n} H(T_n^\pi). \quad (4)$$

Proposition 1 (Policywise operational meaning). *For a fixed policy π , $H(T_n^\pi)$ is the minimum entropy of a random variable E_π from which the exact transcript can be produced deterministically. Consequently, $\rho_n(W)$ is the worst-policy value of this policywise seed cost.*

Proof. If $T_n^\pi = f(E_\pi)$ deterministically, then $H(T_n^\pi) \leq H(E_\pi)$. Equality is achievable by taking $E_\pi = T_n^\pi$ and f to be the identity map. $\square$

This quantity should not be confused with a universal seed sampled before the policy is known. The distinction is already present in one-step entropic causal models: forcing a single exogenous variable to be independent of the input can require more entropy than the conditional output entropy [2].

2.2 Counterfactual common-seed cost

A stronger requirement asks for one policy-independent exogenous variable that fixes responses under all admissible interventions.

Definition 2 (Response tree). *A depth- n response tree is an element $r \in \mathfrak{R}_n$. At each history-action node (h_{t-1}, a_t) it assigns a definite observation in $\mathcal{O}$. Given r and a policy π , recursion through the tree determines a unique transcript, denoted $F_{\pi,n}(r)$.*

Let

$$\mathcal{Q}_n(W) := \{q \in \Delta(\mathfrak{R}_n) : (F_{\pi,n})_{\#} q = \mu_W^{\pi,n} \quad \forall \pi \in \Pi_n\}. \quad (5)$$

For the classical kernel model in Eq. (3), $\mathcal{Q}_n(W)$ is nonempty: one may sample independently, for every history-action node, an output with the corresponding kernel and collect all sampled outputs into a response tree. Because $\mathfrak{R}_n$ is finite, $\mathcal{Q}_n(W)$ is a nonempty compact polytope and Shannon entropy is continuous on it; the minimum below is therefore attained.

Definition 3 (Counterfactual seed cost).

$$\chi_n(W) := \min_{q \in \mathcal{Q}_n(W)} H_q(R_n). \quad (6)$$

Here R_n is the random response tree with law q .

The word *counterfactual* is essential. χ_n charges a single random object that commits to responses on branches that will not all be visited in one run. It is therefore a stronger resource than actual-path randomness and should not be interpreted as the amount of physical randomness necessarily consumed in a single experiment.

Theorem 1 (Operational characterization). *$\chi_n(W)$ is the minimum $H(E)$ over all policy-independent exogenous variables E for which there exists a deterministic causal mechanism*

$$O_t = g_t(H_{t-1}, A_t, E) \quad (7)$$

that reproduces $\mu_W^{\pi,n}$ for every $\pi \in \Pi_n$.

Proof. Fix such a mechanism. Each value e determines the output at every possible history-action node and hence a response tree $R_n = \varphi(E)$. Therefore $H(R_n) \leq H(E)$, and exactness implies that the law of R_n belongs to $\mathcal{Q}_n(W)$. Thus $\chi_n(W) \leq H(E)$. Conversely, if $R_n \sim q \in \mathcal{Q}_n(W)$, take $E = R_n$ and let the deterministic mechanism read the response stored at the currently visited node. Minimizing over q gives equality. $\square$

Proposition 2 (On-path lower bound and counterfactual premium). *For every finite controlled world,*

$$\rho_n(W) \leq \chi_n(W). \quad (8)$$

The inequality can be strict.

Proof. For every $q \in \mathcal{Q}_n(W)$ and every policy π , the transcript $T_n^\pi = F_{\pi,n}(R_n)$ is a deterministic function of R_n . Hence $H(T_n^\pi) \leq H(R_n)$. Take the supremum over π and then the minimum over q .

For strictness, take one time step, actions $a \in \{0, 1\}$, and binary output Y with

$$P(Y = 1 \mid a = 0) = \frac{1}{2}, \quad P(Y = 1 \mid a = 1) = \frac{1}{4}.$$

Then $\rho_1 = 1$ bit. A response tree is a pair (Y_0, Y_1) with Bernoulli marginals $1/2$ and $1/4$. Writing $t = P(Y_0 = 1, Y_1 = 1)$, feasible couplings have $t \in [0, 1/4]$. Entropy is concave in t , so its minimum occurs at an endpoint; both endpoints yield the probability vector $(1/2, 1/4, 1/4, 0)$ up to permutation and hence entropy $3/2$ bits. Thus $\chi_1 = 3/2 > \rho_1$. $\square$

The difference

$$\Delta_n^{\text{cf}}(W) := \chi_n(W) - \rho_n(W) \geq 0$$

may be read as a *counterfactual premium*: the additional entropy required when one common exogenous state must coordinate responses across mutually exclusive interventions. At one step, computing this optimum is precisely a minimum-entropy coupling problem of the kind appearing in entropic causal inference; the general problem is NP-hard [2].

Both costs are nondecreasing with the horizon. For χ_n , truncation of a depth- $(n+1)$ response tree gives an admissible depth- n tree and cannot increase entropy. For ρ_n , any depth- n policy can be extended by an arbitrary final action, while entropy cannot decrease when a transcript is extended. Thus

$$\rho_{n+1} \geq \rho_n, \quad \chi_{n+1} \geq \chi_n. \quad (9)$$

For n independent fair bits with no interventions,

$$\rho_n = \chi_n = n. \quad (10)$$

The law is short to describe, yet an exact realization of the individual sequence carries n bits of Shannon entropy. This ensemble-level statement is conceptually consonant with, but mathematically distinct from, the separation between model information and instance information in algorithmic statistics [8].

3 Resource accounting across realization layers

The preceding definitions support a precise but deliberately narrow resource interpretation. Take random exogenous sources as resources and deterministic, causal, policy-independent processing as free. This is a standard kind of resource-theoretic choice: randomness itself forms a resource under deterministic processing [7]. Here “resource-theoretic” is used in this minimal operational sense: a specified resource, a specified class of free transformations, and monotones under those transformations. We do not claim to develop a complete categorical resource theory with conversion rates, catalysis, or a full set of monotones.

Theorem 2 (Source-aggregation bound). *Suppose an exact common-seed realization of W through horizon n is obtained from exogenous random variables $E_1, \dots, E_m$ such that their joint law is policy-independent and, once they are fixed, all subsequent realization maps are deterministic and causal. Then*

$$\boxed{H(E_1, \dots, E_m) \geq \chi_n(W)}. \quad (11)$$

Consequently,

$$\boxed{\sum_{j=1}^m H(E_j) \geq \chi_n(W)}, \quad (12)$$

with equality between joint entropy and the sum when the sources are independent.

Proof. The joint source $E = (E_1, \dots, E_m)$ deterministically fixes the complete response tree R_n . Hence

$$H(E_1, \dots, E_m) \geq H(R_n) \geq \chi_n(W).$$

Subadditivity gives $H(E_1, \dots, E_m) \leq \sum_j H(E_j)$. $\square$

This theorem is the exact content of the “information debt” metaphor in the common-seed model. Deterministic layers may transform or redistribute the source information; they cannot generate the required joint entropy from a zero-entropy input. The theorem does *not* apply unchanged to a model in which fresh randomness is created only after a particular branch is selected and no policy-independent joint counterfactual source is posited. For such on-demand models, ρ_n is the more conservative lower bound. Keeping these two accounting conventions separate is essential.

4 Admissible host classes and approximation

No purely information-theoretic host class should be confused with the class of physically possible implementations. Let $\mathfrak{A}$ denote a specified class of deterministic common-seed causal realizations. It may encode constraints such as locality, memory architecture, communication restrictions, computability, or a chosen nontrivial criterion of implementation. For finite distributions we use the convention

$$d_{\text{TV}}(P, Q) := \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

For $\varepsilon \geq 0$, define

$$\chi_{n,\varepsilon}^{\mathfrak{A}}(W) := \inf \left\{ H(E) : (G, E) \in \mathfrak{A}, \sup_{\pi \in \Pi_n} d_{\text{TV}}(\mu_{G,E}^{\pi,n}, \mu_W^{\pi,n}) \leq \varepsilon \right\}, \quad (13)$$

with $\inf \emptyset = +\infty$.

Proposition 3 (Monotonicity under restriction and smoothing). *If $\mathfrak{A}_1 \subseteq \mathfrak{A}_2$ and $0 \leq \varepsilon_1 \leq \varepsilon_2$, then*

$$\chi_{n,\varepsilon_1}^{\mathfrak{A}_1}(W) \geq \chi_{n,\varepsilon_2}^{\mathfrak{A}_2}(W). \quad (14)$$

Proof. The feasible set on the left is a subset of the feasible set on the right. $\square$

The dependence on ε can be bounded quantitatively. For the nontrivial case $|\mathcal{O}| \geq 2$, let $M_n := |\mathcal{O}|^n$, an upper bound on the number of transcripts compatible with any fixed deterministic policy through horizon n , and let $h_2(x) := -x \log_2 x - (1-x) \log_2(1-x)$ denote the binary entropy, with the usual continuous convention at 0 and 1.

Proposition 4 (Robust on-path lower bound). *Assume $0 \leq \varepsilon \leq 1 - 1/M_n$. For every admissible class $\mathfrak{A}$ of deterministic common-seed realizations,*

$$\boxed{\chi_{n,\varepsilon}^{\mathfrak{A}}(W) \geq [\rho_n(W) - h_2(\varepsilon) - \varepsilon \log_2(M_n - 1)]_+}. \quad (15)$$

Here $[x]_+ := \max\{x, 0\}$.

Proof. Fix a feasible realization with seed E , and fix a deterministic policy π . Let $P_\pi = \mu_W^{\pi,n}$ be the target transcript law and let Q_π be the law generated by the realization. Under fixed π , the realized transcript is a deterministic function of E , so $H(E) \geq H(Q_\pi)$. Let $\delta_\pi := d_{\text{TV}}(P_\pi, Q_\pi) \leq \varepsilon$. The sharp finite-alphabet continuity bound for Shannon entropy, together with monotonicity of $h_2(\delta) + \delta \log_2(M_n - 1)$ on $0 \leq \delta \leq 1 - 1/M_n$, gives

$$H(Q_\pi) \geq H(P_\pi) - h_2(\varepsilon) - \varepsilon \log_2(M_n - 1).$$

Taking the supremum over π , using the definition of ρ_n , and then taking the infimum over feasible realizations yields Eq. (15); nonnegativity of entropy gives the positive part. The continuity inequality is the classical restriction of the sharp Fannes–Audenaert bound [4]. $\square$

For n independent fair output bits with no interventions, $M_n = 2^n$ and $\rho_n = n$, hence

$$\chi_{n,\varepsilon} \geq n - h_2(\varepsilon) - \varepsilon \log_2(2^n - 1) \geq (1 - \varepsilon)n - h_2(\varepsilon). \quad (16)$$

Thus a fixed small total-variation error still leaves a linear exact-statistical resource requirement. This should be contrasted with computational indistinguishability below, where a short pseudorandom seed can suffice for a restricted distinguisher class.

Thus realistic restrictions do not refute the resource formulation; they parameterize it. They may raise the required seed entropy or make exact realization impossible within the chosen class. What does *not* follow without an additional physical bridge theorem is a universal conversion of $H(E)$ into energy, spatial volume, clock rate, or thermodynamic dissipation. For example, Landauer-type bounds concern logically irreversible erasure under specific thermodynamic assumptions, not arbitrary storage or generation of a seed. Physical resource coordinates therefore have to be added rather than silently identified with χ_n .

The exact requirement is likewise substantive. Exact and total-variation approximate channel synthesis can have different resource regions [5], and sequential synthesis has its own nontrivial optimization structure [6]. Equation (13) makes the accuracy dependence explicit rather than treating exactness as a hidden assumption.

For computationally bounded observers, an even weaker notion is natural. If $\mathcal{D}$ is a class of distinguishers, define

$$d_{\mathcal{D}}(P, Q) := \sup_{D \in \mathcal{D}} \left| \Pr_P[D = 1] - \Pr_Q[D = 1] \right|$$

and replace total variation in Eq. (13) by $d_{\mathcal{D}}$. The resulting cost $\chi_{n,\varepsilon}^{\mathcal{D}}$ satisfies, for the same underlying realization class,

$$\chi_{n,\varepsilon}^{\mathcal{D}} \leq \chi_{n,\varepsilon}^{\text{TV}} \leq \chi_{n,0}. \quad (17)$$

A pseudorandom generator illustrates the distinction. For an m -bit uniform source, exact cost is m bits. If a deterministic generator $G : \{0, 1\}^k \rightarrow \{0, 1\}^m$ with $k < m$ is indistinguishable from uniform for the chosen observer class, then the observer-relative cost is at most k . The missing exact entropy has not been paid; the discrepancy has been made computationally inaccessible. Standard constructions of such generators are connected to one-way functions [9].

5 Infinite horizons: the consistency issue

Finite-horizon minima do not by themselves establish a single infinite realization that is simultaneously optimal at every horizon. This is a real issue and should not be hidden inside the notation χ_n .

Let $\mathfrak{R}_\infty$ be the space of response assignments on all finite history–action nodes, and let $\mathcal{Q}_\infty(W)$ consist of probability measures q on $\mathfrak{R}_\infty$ whose depth- n marginal belongs to $\mathcal{Q}_n(W)$ for every n . In the finite-alphabet kernel model of Eq. (3), this set is nonempty: sampling every node independently from its corresponding kernel defines such a measure. In more general theories of intervention processes, extension and consistency require additional care; generalized Kolmogorov extension results provide the relevant framework [11].

Define the finite-horizon lower rate and the single-realization global rate by

$$\bar{\chi}_{\text{fin}}(W) := \limsup_{n \rightarrow \infty} \frac{\chi_n(W)}{n}, \quad (18)$$

$$\chi_{\text{glob}}(W) := \inf_{q \in \mathcal{Q}_\infty(W)} \limsup_{n \rightarrow \infty} \frac{H_q(R_n)}{n}. \quad (19)$$

Proposition 5 (Finite-horizon lower bound on global rate).

$$\boxed{\chi_{\text{glob}}(W) \geq \bar{\chi}_{\text{fin}}(W)}. \quad (20)$$

Proof. For every $q \in \mathcal{Q}_\infty(W)$, its depth- n marginal lies in $\mathcal{Q}_n(W)$, so $H_q(R_n) \geq \chi_n(W)$ for every n . Divide by n , take lim sup, and then the infimum over q . $\square$

No equality is claimed, and no ordinary limit is assumed to exist. Moreover, $\chi_n > 0$ at every finite horizon does not imply a positive asymptotic rate: the sequence may be bounded or sublinear. Claims about a persistent per-time “debt” therefore require an explicit asymptotic statement such as $\bar{\chi}_{\text{fin}} > 0$ or $\chi_{\text{glob}} > 0$. The normalization by n measures cost per unit temporal horizon; for strongly branching counterfactual trees, a normalization by the number of decision nodes may be appropriate for a different operational question. No universal physical meaning is attached to either normalization. For independent fair bits without interventions, both rates above equal one bit per step.

6 What the bounds say about the substrate

The key observation is that ρ_n and χ_n are functionals of the internal policy-indexed interface, not identifiers of its realization.

Theorem 3 (Substrate invariance of the interface costs). *If $R_1 \equiv_{\text{obs}} R_2$, then for every finite horizon n ,*

$$\boxed{\rho_n(R_1) = \rho_n(R_2), \quad \chi_n(R_1) = \chi_n(R_2)}. \quad (21)$$

Whenever the corresponding infinite-horizon interfaces are defined, the rates in Eqs. (18)–(19) are likewise equal.

Proof. Observational equivalence gives identical transcript laws for every policy and horizon. Therefore the suprema defining ρ_n coincide, the feasible response-tree sets $\mathcal{Q}_n$ coincide, and hence their entropy minima coincide. The rate statement follows horizon by horizon. $\square$

Thus resource information can refine substrate skepticism without defeating it. If $\chi_n(W) = r$, then a deterministic common-seed realization with less than r bits of joint exogenous entropy is impossible. If a physically motivated admissible class $\mathfrak{A}$ yields $\chi_{n,0}^{\mathfrak{A}} = +\infty$, that entire class is excluded. These are real constraints. But they do not select one member of an observational equivalence class, and they do not determine where the required exogenous information is supplied.

There is also an abstract exact lifting construction.

Proposition 6 (Abstract exact lifting). *For every finite W and n , there is an abstract external causal generator that samples an optimal response tree with entropy $\chi_n(W)$ and thereafter reproduces the complete depth- n policy-indexed interface deterministically.*

Proof. Choose $q^* \in \mathcal{Q}_n(W)$ attaining the minimum, sample $R_n \sim q^*$, and use the deterministic response-tree interpreter. $\square$

A slightly stronger consequence is available, but only relative to an implementation class that counts observationally silent wrappers as distinct realization layers.

Corollary 1 (No finite depth bound under wrapper closure). *Suppose an admissible realization class contains the exact lifted generator of Proposition 6 and is closed under insertion of deterministic, causal wrappers whose composition preserves the same policy-indexed interface. If each inserted wrapper is counted by the chosen implementation criterion as a distinct realization layer, then for every $m \in \mathbb{N}$ there exists an observationally equivalent realization chain with at least m such layers. Hence the internal interface alone supplies no finite upper bound on realization depth within that class.*

Proof. Start from the lifted generator and insert m deterministic identity-equivalent wrappers between the seed-bearing generator and the internal interface. By closure, the resulting chain is admissible. Its composite input–output map is unchanged, so all policy-indexed transcript laws are unchanged. $\square$

The closure assumption is substantive. Proposition 1 is not a claim that arbitrary software wrappers constitute distinct physical worlds; it states exactly why no depth conclusion follows until a nontrivial theory of physical implementation says which realization layers count.

This is a theorem about causal generation, not a theorem that the generator is a physically possible host. If a proposed physical implementation criterion admits such a generator, positive χ_n plainly does not certify fundamentality. If it does not, the obstruction comes from the additional physical criterion, not from χ_n alone. This distinction is important in light of implementation-trivialization worries: preserving an entire family of counterfactual responses is stronger than retrospectively relabeling one actual trajectory, but it is not by itself a complete metaphysics of physical computation [12].

Likewise, observable substrate traces do not contradict substrate skepticism. If thermodynamic, quantum, cosmological, or other effects alter an internally accessible transcript distribution, then Eq. (2) simply fails and the relevant realizations are no longer observationally equivalent. Substrate skepticism is not the empirical thesis that hosts leave no traces; it is the conditional claim that realization status is not recoverable *within an observational-equivalence class*.

7 From information debt to simulation no-go and fundamentality

The preceding bounds constrain realizations, but different stronger conclusions require different additional premises. In particular, the statement that a world is not exactly simulable by a specified host class is substantially weaker than the statement that stochasticity is ontically fundamental. The first can already be obtained from a finite information budget; the second requires a bridge from information accounting to the ontology of the terminal layer.

7.1 A relative no-go theorem for closed deterministic hosts

Call a realization *closed deterministic through horizon n* if there is a policy-independent initial random variable Z such that, once Z is fixed, all responses to all admissible interventions are generated deterministically and no additional exogenous random variable is supplied during the run. Such a Z is a common seed in the sense of Theorem 1.

Theorem 4 (Closed finite-seed no-go). *Suppose W is exactly realized through horizon n by a closed deterministic host with policy-independent initial source Z . Then*

$$\boxed{H(Z) \geq \chi_n(W) \geq \rho_n(W)}. \quad (22)$$

Consequently, if $H(Z) \leq B$ and $\chi_n(W) > B$, no such exact realization exists through horizon n . If $\sup_n \chi_n(W) = \infty$, no single closed deterministic host with finite $H(Z)$ realizes the complete interface at all finite horizons.

Proof. For fixed $Z = z$, the deterministic host fixes the response at every admissible history–action node, hence determines a response tree $R_n = \varphi_n(Z)$. Therefore $H(Z) \geq H(R_n) \geq \chi_n(W)$. Proposition 2 gives $\chi_n(W) \geq \rho_n(W)$. The remaining statements are immediate. $\square$

The theorem is relative to this host class: unbounded boundary information, fresh exogenous input, or a non-deterministic host remain open alternatives. By Proposition 4, if $H(Z) \leq B$ and

$$B < [\rho_n(W) - h_2(\varepsilon) - \varepsilon \log_2(M_n - 1)]_+, \quad (23)$$

then no realization in that class can remain uniformly within ε total variation through horizon n . Small statistical error therefore weakens rather than discontinuously removes the obstruction.

7.2 Stored information versus continuing innovation

At infinite horizon there is a second distinction. A deterministic terminal layer can support an indefinitely stochastic-looking target either by carrying an unbounded amount of relevant information in its boundary data or by receiving additional exogenous information as time proceeds. Information theory alone does not identify these two mechanisms.

Let Z denote the terminal initial or boundary source and let $U_{1:n} = (U_1, \dots, U_n)$ collect all additional exogenous variables made available through time n . Assume that, conditional on $(Z, U_{1:n})$, the realized transcript under each deterministic policy is generated deterministically.

Proposition 7 (Stored-information–innovation alternative). *For every n ,*

$$\boxed{H(Z, U_{1:n}) \geq \rho_n(W)}. \quad (24)$$

Hence

$$H(U_{1:n} \mid Z) \geq \rho_n(W) - H(Z). \quad (25)$$

If

$$\underline{\rho}(W) := \liminf_{n \rightarrow \infty} \frac{\rho_n(W)}{n} = h > 0 \quad (26)$$

and $H(Z) < \infty$, then

$$\boxed{\liminf_{n \rightarrow \infty} \frac{H(U_{1:n} \mid Z)}{n} \geq h}. \quad (27)$$

Thus a finite-entropy boundary source cannot by itself support a target with positive asymptotic on-path entropy rate: a positive asymptotic rate of additional exogenous information is required.

Proof. For any fixed policy, the transcript is a deterministic function of $(Z, U_{1:n})$, so $H(Z, U_{1:n}) \geq H(T_n^\pi)$. Taking the supremum over policies gives Eq. (24). Since $H(Z, U_{1:n}) = H(Z) + H(U_{1:n} \mid Z)$, Eq. (25) follows. Divide by n and use $H(Z)/n \rightarrow 0$ to obtain Eq. (27). $\square$

If a single policy-independent source collection fixes the entire counterfactual response tree, the stronger common-seed accounting of Theorem 2 applies and χ_n can replace the on-path lower bound. Proposition 7 deliberately uses ρ_n because it remains valid under the weaker assumption that fresh exogenous information need only support the branch actually realized.

Thus persistent exact stochasticity can be supported either by *stored particularity* in unbounded boundary data or by *continuing innovation*; positive output entropy rate alone does not distinguish them.

7.3 What additional assumptions would imply fundamental stochasticity?

To infer fundamental stochasticity rather than merely a resource requirement, one must add premises that rule out both indefinite descent and unlimited stored boundary information. For a chosen realization theory, consider:

- (I) **Terminality (T)**: every admissible realization hierarchy starting from W reaches a terminal layer after finitely many steps;
- (II) **Resource ancestry (RA)**: every exogenous source at a nonterminal layer is itself physically accounted for by deeper-layer states or sources, and conditional on the complete accounted source collection the inter-layer realization maps are deterministic and causal;
- (III) **Finite boundary information (FB)**: the terminal initial or boundary source Z available to the relevant realization has $H(Z) < \infty$ (or, in a stronger algorithmic formulation, does not contain an unbounded noncomputable “random tape”);
- (IV) **Ontic bridge (OB)**: positive irreducible conditional entropy rate in the terminal exogenous innovation process represents ontic stochasticity rather than merely epistemic uncertainty over a more complete deterministic state.

Theorem 5 (Conditional fundamentality of stochastic innovation). *Assume $\underline{\rho}(W) = h > 0$ and assumptions (T), (RA), (FB), and (OB). Then every exact realization hierarchy of W contains, at a terminal layer, a fundamental ontic stochastic resource with asymptotic exogenous information rate at least h in the sense of Eq. (27).*

Proof. By (T), source ancestry cannot continue indefinitely. By (RA), exogenous resources introduced above the terminal layer are charged to deeper resources rather than being declared primitive there. At the terminal layer, (FB) prevents the initial source Z from carrying a linearly growing amount of relevant entropy. Proposition 7 therefore requires additional terminal exogenous information with asymptotic conditional entropy rate at least h . By (OB), this residual innovation is interpreted as ontic stochasticity. Since the layer is terminal under the chosen realization theory, that stochastic resource is fundamental relative to that theory. $\square$

Each bridge assumption blocks a distinct countermodel: without (T) there may be infinite regress; without (RA) accounting may stop prematurely; without (FB) unbounded boundary data may carry the future information; without (OB) positive Shannon entropy may remain merely epistemic. Accordingly,

$$\boxed{\text{resource necessity} \not\Rightarrow \text{resource localization} \not\Rightarrow \text{fundamentality}.} \quad (28)$$

The missing arrows are supplied only by additional physical or metaphysical premises.

7.4 Fundamental randomness is not a fundamental target world

Even Theorem 5 does not imply $\text{Fund}(W)$: a nonfundamental host may transmit a fundamental host-level stochastic resource across the realization boundary while preserving the target interface. Thus

$$\boxed{\text{fundamental stochastic resource} \not\Rightarrow \text{Fund}(W)}. \quad (29)$$

Fundamental randomness in the provenance chain is therefore compatible with a nonfundamental target world. Likewise, a universal anti-simulation conclusion would require the further premise that every physically possible host belongs to an excluded class; information theory alone does not provide that premise.

The resulting interpretation is concise: exact stochasticity constrains the *provenance* of information more directly than it constrains realization status. Information can cross a simulation boundary without making that boundary internally observable.

8 Relation to semantic compression

The motivating intuition can be stated without putting it into the theorem statements. A stochastic law may have a short semantic or algorithmic description while a particular exact realization carries substantial instance information. Algorithmic statistics formalizes a related distinction through two-part descriptions: a model encoding regularity and an additional code locating the individual datum within the model [8]. This is a useful analogy for the informal language of “harmony” and “particularity.”

It should not, however, be conflated with the present resource measure. χ_n and ρ_n are Shannon entropies of ensemble-valued exogenous resources. Kolmogorov complexity is an individual-object description length and is generally uncomputable. In particular, the existence of a short deeper description does not imply that the description can be found efficiently. Computationally hidden structure is precisely why the exact and observer-relative costs in Eq. (17) must remain distinct. Algorithmic no-randomness-from-nothing results provide a complementary infinite-sequence perspective under specified effective measure-preserving maps, but they do not by themselves establish an ontological source of physical randomness [10].

9 Conclusion

The resource-theoretic refinement of substrate skepticism is narrower than the claim that stochasticity reveals a simulation and stronger than the claim that nothing can be said about a substrate from within. Two exact costs must first be separated. The on-path cost

$$\rho_n = \sup_{\pi} H(T_n^{\pi})$$

measures the worst policywise entropy of what actually occurs. The stronger counterfactual cost

$$\chi_n = \min_{q \in \mathcal{Q}_n} H_q(R_n)$$

measures the entropy of one policy-independent seed that fixes all admissible counterfactual responses. Their difference is a genuine counterfactual premium, not randomness consumed on a single path.

For common-seed realizations, exact stochasticity therefore imposes a quantifiable information requirement: jointly supplied exogenous sources must carry at least χ_n bits of Shannon entropy, and deterministic realization layers cannot create that entropy from nothing. Restricting the admissible host class can only raise the cost or make realization impossible. Allowing a small total-variation error lowers the operational requirement only within the explicit continuity allowance of Proposition 4; restricting the observer computationally can lower it much further. Infinite-horizon claims require a separate consistency and rate analysis.

None of these facts identifies the realization. The costs are invariants of the internal intervention family and are therefore shared by observationally equivalent realizations. Theorem 4 yields a genuine but relative no-go result: a closed deterministic host with a bounded-entropy initial source cannot realize indefinitely a target whose exact information requirements grow without bound. Proposition 7 then separates two remaining possibilities—unbounded stored boundary information and continuing exogenous innovation. Only after adding terminality, resource ancestry, finite boundary information, and an ontic bridge does Theorem 5 turn a positive asymptotic on-path rate into a claim of fundamental stochastic innovation. Even then, Eq. (29) shows that a fundamental stochastic source does not imply that the target world itself is fundamental. The resulting position can be summarized as

Exact stochastic structure can constrain the provenance and minimum information resources of specified realization classes without identifying the actual realization or establishing that the target world is fundamental.	(30)
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In this precise sense, substrate skepticism survives but ceases to treat the substrate as an unconstrained black box: *exact stochasticity constrains the provenance of information more strongly than it constrains realization status. The realization may remain unidentified even when some of the resources required to realize it do not.*

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